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Investigations on a 3.3 K four-stage Stirling-type pulse tube cryocooler. Part B: Experimental verifications

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ABSTRACT

In this part, a four-stage Stirling-type pulse tube cryocooler (SPTC) is worked out to verify the theoretical analyses conducted in Part A. Experiments are carried out to investigate the interactions among the four stages. The operating frequency, charge pressure and input power of the last two stages are optimized with He-4 and He-3, respectively. The experimental results show that a no-load temperature of 4.2 K is reached with He-4 and it further decreases to 3.3 K with He-3. The ability to providing simultaneous cooling powers is also tested. With He-3, the cooling powers of 4.4 W at 70 K, 1.0 W at 40 K, 0.29 W at 15 K and 0.025 W at 5 K are simultaneously achieved by the four stages, respectively. Satisfactory agreements are observed between theoretical and experimental studies.

1. Introduction

The systematic theoretical analyses have been conducted in Part A [1], in which a CFD model is built for the four-stage Stirling-type pulse tube cryocooler (SPTC). The loss mechanisms and phase characteristics in the regenerator are simulated, and the complicated interactions among the four stages are clarified. The theoretical studies reveal the operating mechanisms in the fourth stage cold finger and also provide a systematic optimization for the four-stage SPTC to help improve the cooling performance. In this part, the experimental verifications of the developed four-stage SPTC will be presented, and the simulated and experimental results will be compared and discussed.

2. Experimental verifications

2.1. Experimental setups

To verify the validity of the theoretical studies, a four-stage SPTC is then worked out and tested. A schematic of the four-stage SPTC is shown in Fig. 1. It is driven by two linear compressors developed in the same laboratory [2,3], in which the first two stages are driven by one compressor and the last two stages by the other one. The four stages are thermally coupled by external copper straps. The first three stages have coaxial arrangements while the fourth stage is in-line.

Fig. 2 shows a photo of the actual four-stage CF, and Fig. 3 provides its experimental setup, in which the phase-shifter of the fourth stage is cooled by the third stage CHX. Normally, to study the effect of the phase-shifter temperature, an additional cooler is usually employed to control it. However, for a self-precooled SPTC like the one studied here, it is difficult to control the phase-shifter temperature with other conditions kept the same. Or alternatively, a heat source is used to control the phase-shifter temperature, but it would bring extra heat load to the third stage CHX and thus increase the cooling temperature, which will make the effect of phase-shifter temperature on the performance of the four stages unclear. Therefore, during the test, the phase-shifter temperature is in keeping with the cooling temperature at the third stage.

First of all, the cooling performance of the first two stages is tested separately before attached to the last two ones. Fig. 4 shows the variations of no-load temperatures and cooling powers of the two stages with the input power of compressor I. With a charge pressure of 3.3 MPa and an operating frequency of 55 Hz, the maximum input power of compressor I is 400 W. The no-load temperatures of the two stages are 46.0 K and 15.8 K, respectively. When W_1 is above 350 W, Q_{c1} and Q_{c2} increase with the increasing W_1 but the growth tends to slow. Two cooling powers of 8.0 W at 70 K and 4.6 W at 40 K are achieved with the maximum input power of 400 W.

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Nomenclature

f_1	frequency of compressor I
f_2	frequency of compressor II
p_1	charge pressure of compressor I
p_2	charge pressure of compressor II
Q	cooling power
T	temperature
W_1	input power of compressor I
W_2	input power of compressor II

Greeks

η	relative Carnot efficiency

Subscripts

1	first stage
2	second stage
3	third stage
4	fourth stage
c	cold end
p	precooling power
p_s	phase-shifter

2.2. He-4 in the last two stages

After matching the first two stages to the last two stages, T_{c1} and T_{c2} are fixed at 70 K and 40 K, respectively, which are controlled by heater bands attached to the CHXs. Then the values of Q_{c1} and Q_{c2} can be acquired.

Fig. 5 shows the variations of T_{c3} and T_{c4} with f_2 and p_2, respectively. In the third stage, when p_2 increases from 1.0 MPa to 1.2 MPa and then to 1.4 MPa, the minimum value of T_{c3} increases from 12.7 K to 12.8 K and then to 12.86 K, accordingly the optimal frequencies are 33.0 Hz, 32.2 Hz and 30.8 Hz, respectively. In the fourth stage, the minimum values of T_{c4} at the three pressures are 4.2 K, 4.3 K and 4.6 K, respectively, accordingly the optimal frequencies are 31.2 Hz, 30.5 Hz and 29.6 Hz, respectively. According to the simulated results in Fig. 23 in Part A [1], the optimal frequency of the third stage is higher than that of the fourth stage, and it decreases with the increasing charge pressure, which are in good agreement with the experimental ones.

Fig. 6 shows the variations of Q_{c3} and Q_{c4} with f_2 and p_2, while T_{c3} and T_{c4} are fixed at 15 K and 5 K, respectively. In the third stage, the maximum values of the three solid curves are very close. In the fourth stage, when p_2 is 1.2 MPa and f_2 is 30.5 Hz, Q_{c4} gets the maximum value of 0.015 W. For the four-stage SPTC aiming at simultaneously providing the cooling powers at the four stages, one principle is that the cooling performance of the lower stage should be prioritized, and the slight reduction of the cooling power at the upper stage is acceptable. Therefore, f_2 and p_2 are fixed at 30.5 Hz and 1.2 MPa, respectively. According to the simulated results in Section 3.2 of Part A [1], the optimal f_2 and p_2 are determined to be 30 Hz and 1.2 MPa, respectively, with He-4, which are in good agreement with the experimental ones.

Fig. 7 shows the effect of W_1 on the no-load temperatures of the four stages. When W_1 increases from 100 W to 400 W, T_{c1} decreases from 104.5 K to 61.2 K, T_{c2} from 68.1 K to 35.0 K, T_{c3} from 24.4 K to 13.3 K, and T_{c4} from 8.80 K to 4.25 K, respectively. When W_1 is above 350 W, the decreases of the four temperatures all become very small.

Fig. 8 presents the variations of cooling powers at four stages. Q_{c1},

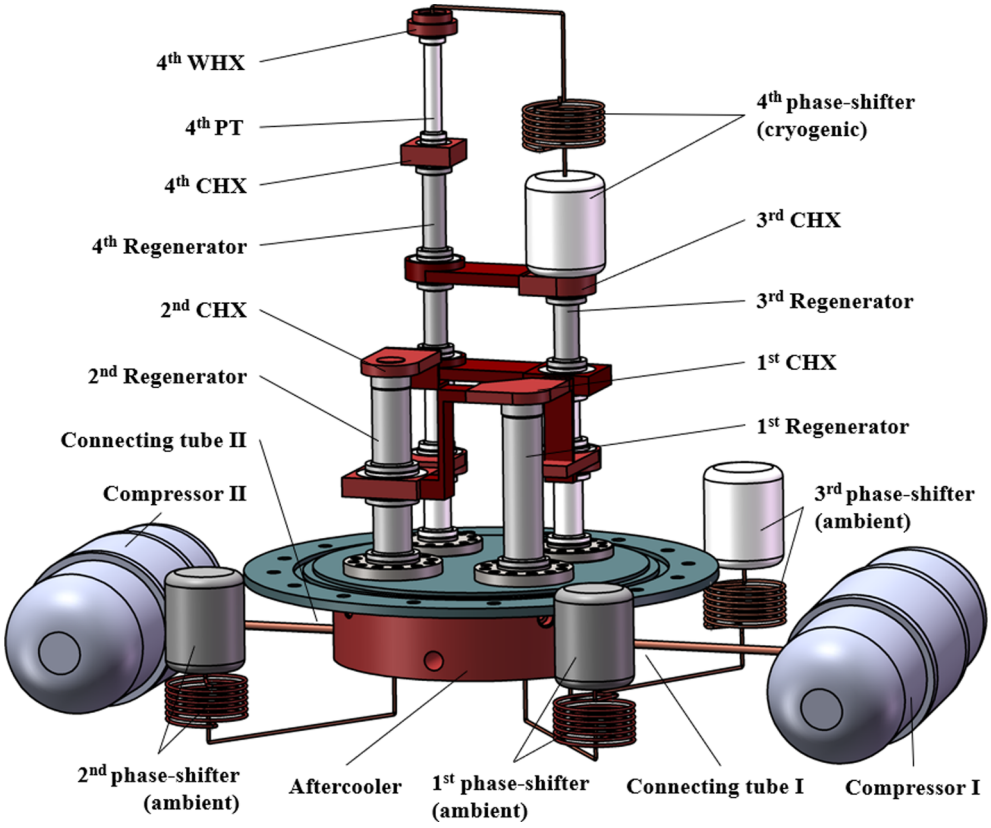


Fig. 1. Schematic of the four-stage SPTC.

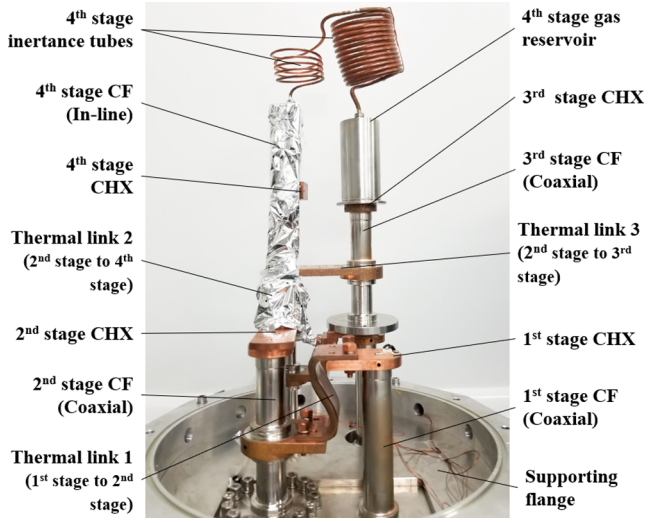


Fig. 2. Photo of the actual four-stage SPTC CF.

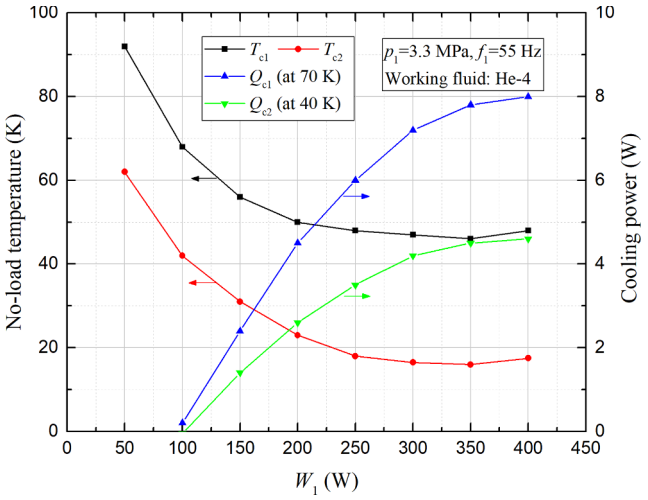


Fig. 4. Effect of W_1 on the cooling performances of the first two stages.

Q_{c2}, Q_{c3} and Q_{c4} all increase with the increasing W_1. When W_1 is above 350 W, the cooling powers of the four stages increase very slowly. Thus, W_1 is determined to be 350 W.

Fig. 9 shows the effect of W_2 on the no-load temperatures of the four stages, respectively. The results are quite different from those in Fig. 7. The maximum input power of compressor II is 140 W when the charge pressure is 1.2 MPa and the operating frequency is 30.5 Hz. When W_2 increases from 10 W to 140 W, both T_{c1} and T_{c2} increase but more and more slowly. The reason is that the more power is put into the last two stages, the more precooling powers will be required from the first two stages. T_{c3} and T_{c4} first decrease and then increase slightly with the increasing W_2. The optimal W_2 for T_{c3} and T_{c4} are 90 W and 100 W, respectively. This is probably because when W_2 is above 90 W, increasing W_2 leads to a limited increment of the cooling power at the third stage but does increase the precooling powers from the third stage to the fourth stage (Q_{p34} and Q_{p4}), which deteriorate the cooling performance of the third stage and then that of the fourth stage as well.

Comparing to the simulated results, the tested T_{c3} has a similar varying trend. The simulated T_{c4} keeps constant when W_2 is above 100 W while the tested T_{c4} increases. The reason might be that the power distributions into

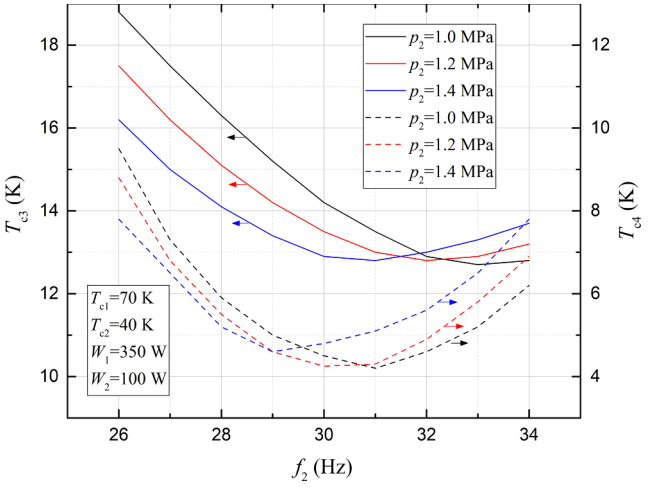


Fig. 5. Variations of T_{c3} and T_{c4} with f_2 and p_2 by using He-4.

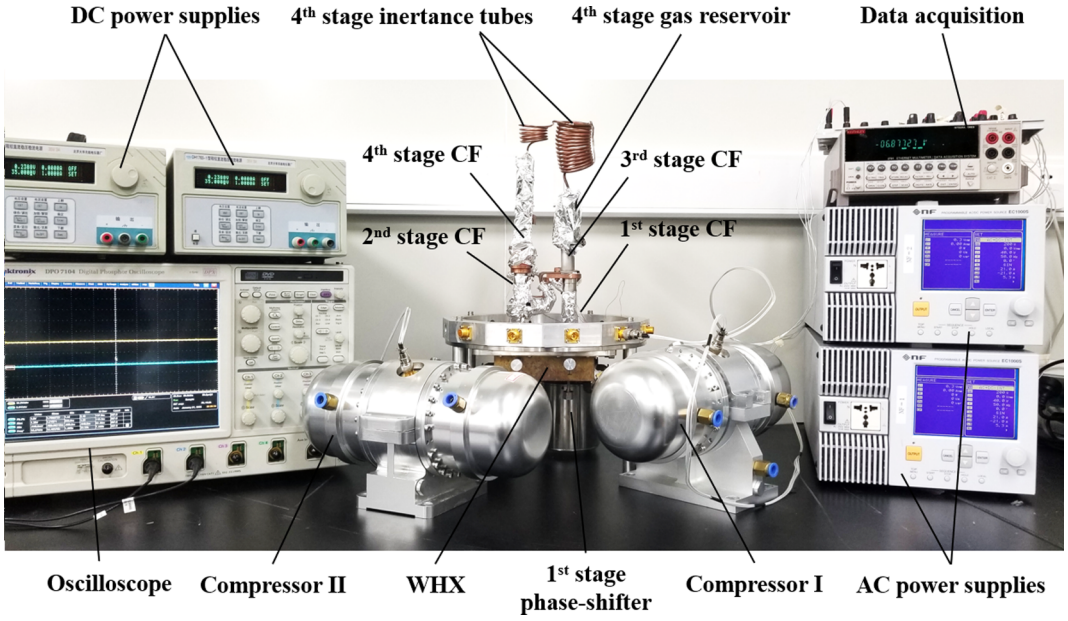


Fig. 3. Experimental setup.

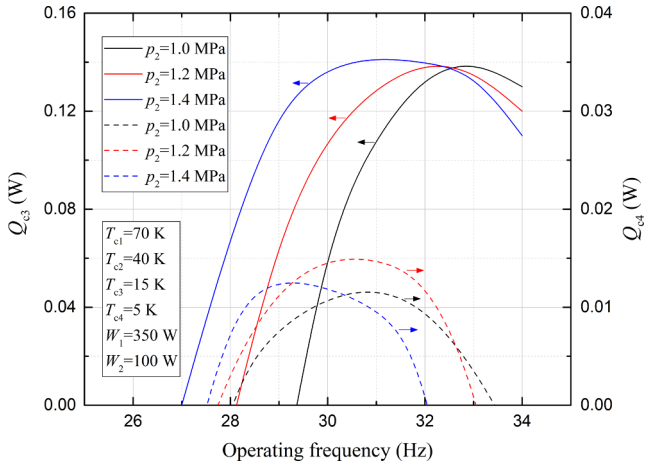


Fig. 6. Variations of Q_{c3} and Q_{c4} with f_2 and p_2 by using He-4.

the third and the fourth stage are different between simulations and experiments. The experimental ratio of the power into the fourth stage to W_2 is larger than the theoretical one. In the simulations, the increase of input power into the fourth stage generates more cooling power and offsets the reduction of performance caused by the increasing T_{c3}. But in experiments, if the power distribution is different from the simulated one, the varying trend of T_{c4} might also be different. The results also indicate that to find the optimal input power is crucial for the cooling performance of the four-stage SPTC.

Fig. 10 presents the variations of cooling powers at four stages. With the increasing W_2, Q_{c1} decreases from 6.8 W to 3.6 W while Q_{c2} from 3.7 W to 0.6 W. Q_{c3} increases from zero to 0.105 W and then decreases to 0.07 W. Q_{c4} increases from zero to 0.015 W and then decrease to 0.002 W. When W_2 is 100 W, Q_{c4} reaches its maximum value while Q_{c3} is slightly smaller than its own maximum value, which is a satisfactory performance of the last two stages. Therefore, W_2 is fixed to be 100 W.

Comparing to the simulated results, the tested Q_{c3} and Q_{c4} have different varying trends. When W_2 is above 90 W, the simulated Q_{c3} keeps increasing but the experimental one begins to decrease. Similarly, when W_2 is above 100 W, the simulated Q_{c4} keeps increasing while the experimental one decreases. The reason is the same as the analyses about the results in Fig. 9.

The experiments are carried out to investigate the cooling performances of the four stages with a varying W_2 with the sum of W_1 and W_2 kept constant, as shown in Fig. 11. W_2 ranges from 50 W to 140 W while W_1 varies from 400 W to 310 W. When W_2 increases from 50 W to

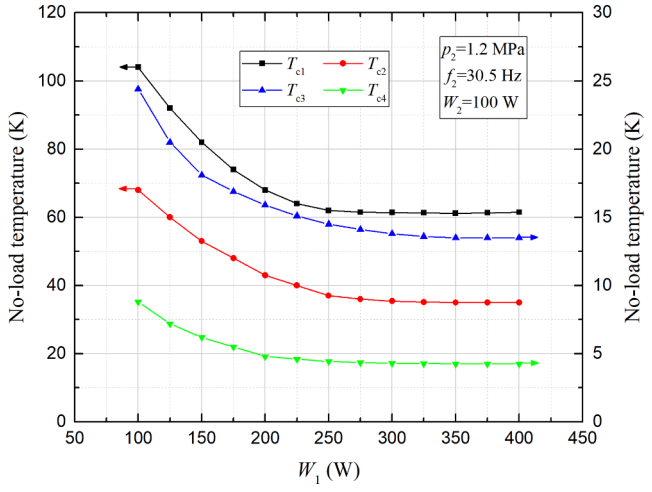


Fig. 7. Effect of W_1 on the no-load temperatures of the four stages.

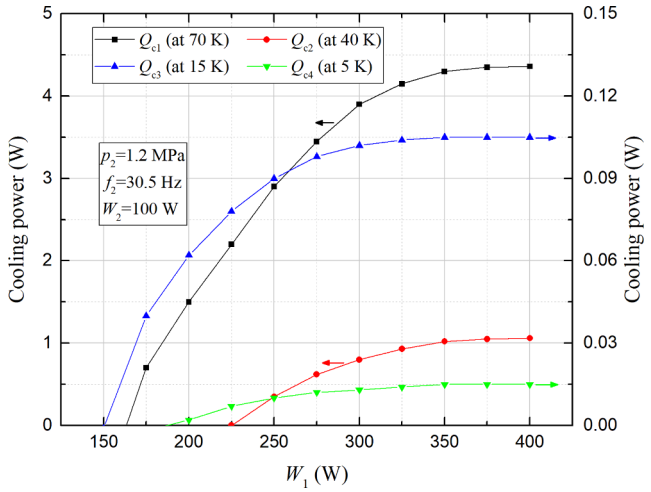


Fig. 8. Effect of W_1 on the cooling powers of the four stages.

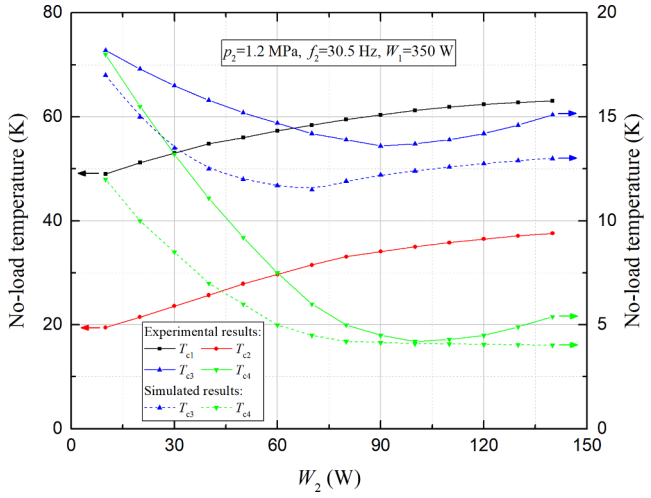


Fig. 9. Effect of W_2 on the no-load temperatures of the four stages.

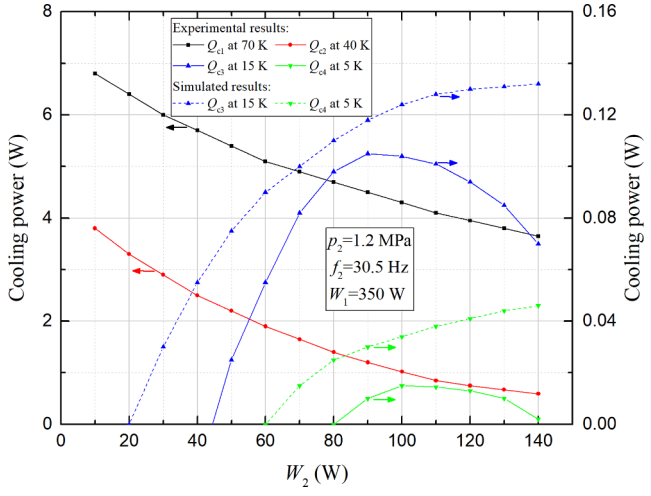


Fig. 10. Effect of W_2 on the cooling powers of the four stages.

140 W, Q_{c1} decreases from 6.0 W to 3.2 W and Q_{c2} from 2.3 W to 0.3 W. Both Q_{c3} and Q_{c4} first increase and then decrease with the increasing W_2, and the optimal W_2 is 100 W. Thus, W_2 is determined to be 100 W.

After the operating frequency, charge pressure and input powers are determined, the interactions between the cooling performance of one

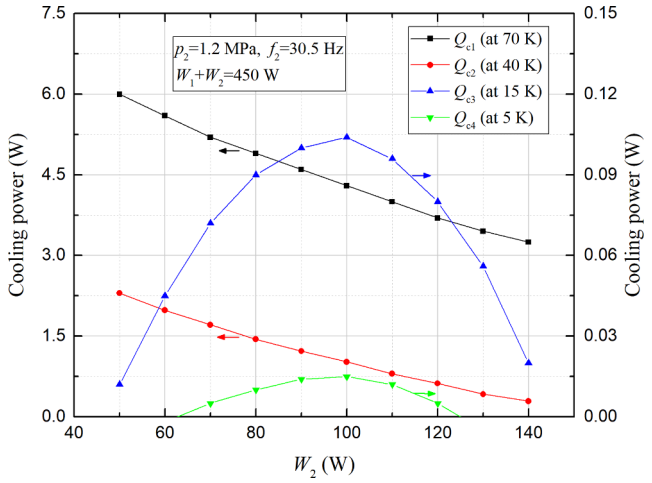


Fig. 11. Effect of W_2 on the cooling powers of the four stages with a total power of 450 W.

stage and those of other three ones will then be investigated.

Fig. 12(a) shows the effect of the first stage cooling temperature on the no-load temperatures of the other three stages. T_{c1} is adjusted by the temperature controller. When T_{c1} increases from 62 K to 80 K, T_{c2} increases from 35.8 K to 43.0 K, T_{c3} from 13.8 K to 15.6 K and T_{c4} from 4.26 K to 5.30 K. T_{c1} has relatively larger influence on T_{c2} than either T_{c3} or T_{c4}. Fig. 12(b) shows the variations of the four cooling powers with T_{c1}. With the increasing T_{c1}, Q_{c1} increases from zero to 9.8 W. With no load in the first stage, Q_{c2}, Q_{c3} and Q_{c4} are 1.1 W, 0.14 W and 0.018 W, respectively. When T_{c1} increases to 73 K, Q_{c2} decreases to zero. When T_{c1} is above 76 K, no cooling powers can be obtained from the third and fourth stages. The varying trends of the simulated Q_{c3} and Q_{c4} are verified by the experiments conducted here.

Fig. 13(a) shows the effect of T_{c2} on the no-load temperatures of the other three stages. When T_{c2} increases from 36 K to 45 K, the increase of T_{c1} is less than 0.1 K while T_{c3} increases from 13.3 K to 16.1 K and T_{c4} from 4.24 K to 5.21 K. The results indicate that T_{c2} has much smaller influence on T_{c1} than on T_{c3} or T_{c4}. Fig. 13(b) shows the variations of the four cooling powers with T_{c2}. With the increasing T_{c2}, Q_{c1} almost keeps unchanged and Q_{c2} increases from zero to 2.9 W. With no load at the second stage, Q_{c1}, Q_{c3} and Q_{c4} are 4.2 W, 0.17 W and 0.02 W, respectively. When T_{c2} is above 41.7 K, no cooling powers can be

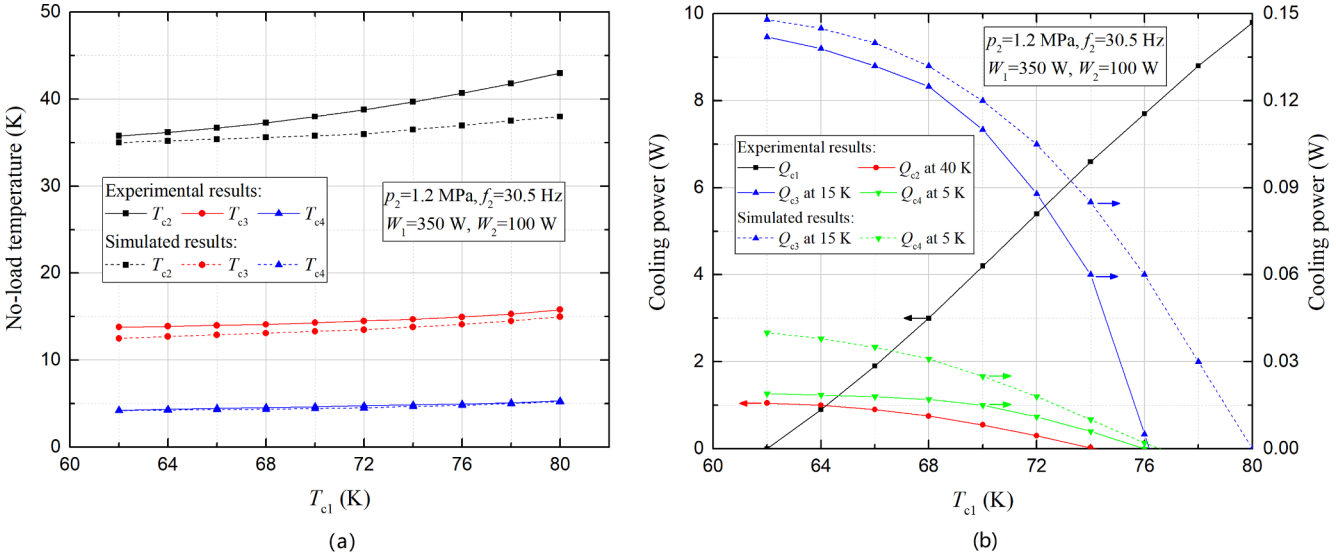


Fig. 12. Effect of T_{c1} on (a) T_{c2}, T_{c3} and T_{c4}, (b) Q_{c1}, Q_{c2}, Q_{c3} and Q_{c4}.

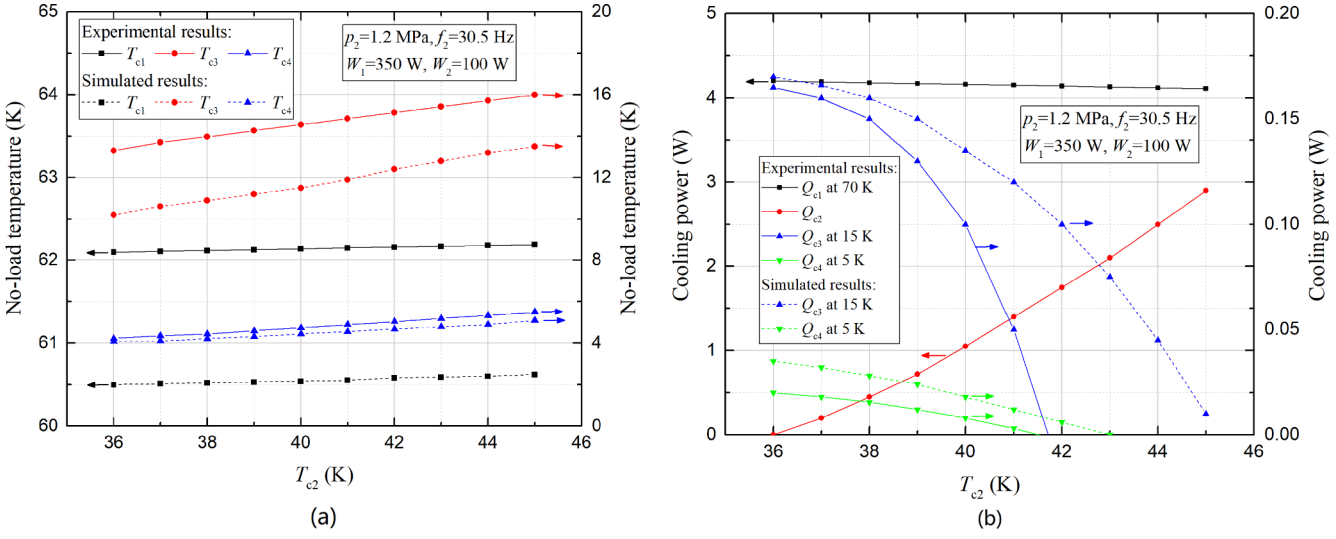


Fig. 13. Effect of T_{c2} on (a) T_{c1}, T_{c3} and T_{c4}, (b) Q_{c1}, Q_{c2}, Q_{c3} and Q_{c4}.

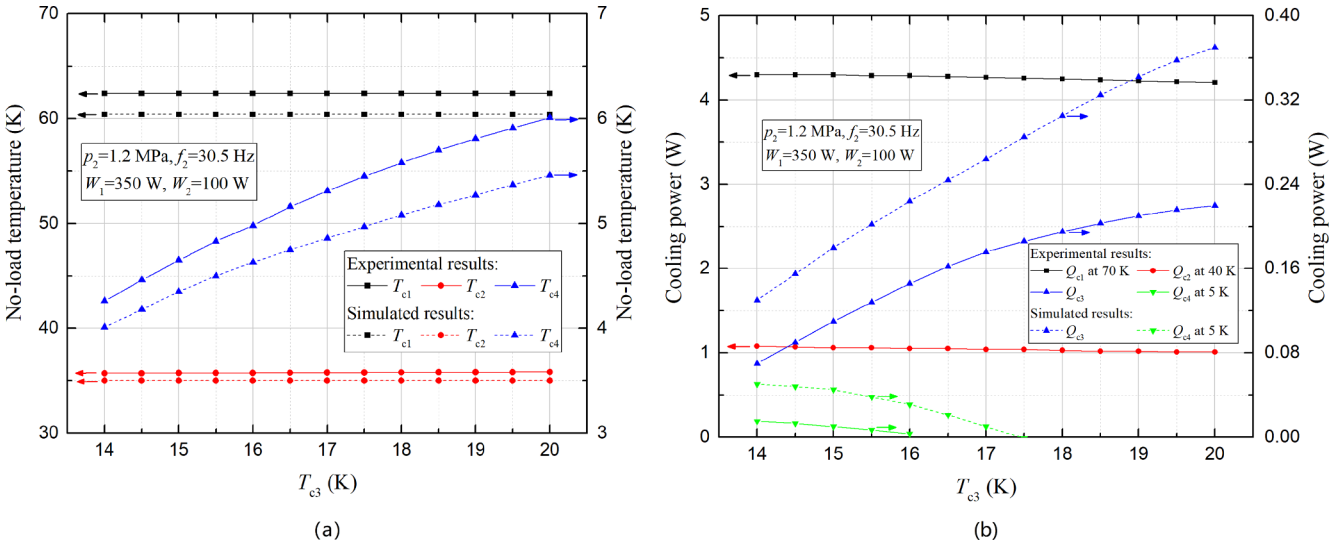


Fig. 14. Effect of T_{c3} on (a) T_{c1}, T_{c2} and T_{c4}, (b) Q_{c1}, Q_{c2}, Q_{c3} and Q_{c4}.

obtained from the third and fourth stages. The above experiments verify the varying trends of the simulated results of Q_{c3} and Q_{c4}.

Fig. 14(a) shows the effect of T_{c3} on the no-load temperatures of the other three stages. When T_{c3} increases from 14 K to 20 K, T_{c1} and T_{c2} almost keeps unchanged while T_{c4} increases from 4.23 K to 6.02 K. The results indicate that T_{c3} affects T_{c4} greatly but has little influence on T_{c1} and T_{c2}. Fig. 14(b) shows the variations of the four cooling powers with T_{c3}. With the increasing T_{c3}, Q_{c1} and Q_{c2} almost keeps unchanged, while Q_{c3} increases from 0.07 W to 0.22 W. When T_{c3} is above 16 K, no cooling power is obtained from the fourth stages. These experimental results have the same varying trends as those of the simulated results.

2.3. He-3 in the last two stages

In order to further improve the cooling performances of the SPTC, He-3 is used to replace He-4 in the last two stages. According to the simulated results in Part A [1], the optimal operating parameters of compressor II with He-3 are different from those with He-4. Therefore, similar experiments are carried out to find the optimal operating frequency and charge pressure.

Fig. 15 shows the variations of T_{c3} and T_{c4} with f_2 and p_2. In the third stage, when p_2 increases from 0.8 MPa to 1.0 MPa and then to 1.2 MPa, the minimum value of T_{c3} varies from 9.5 K to 9.6 K and then to 10.0 K, accordingly the optimal frequencies are 34.8 Hz, 33.9 Hz and 33.5 Hz, respectively. In the fourth stage, the minimum values of T_{c4} at the three pressures are 3.25 K, 3.30 K and 3.56 K, accordingly the optimal frequencies are 34.1 Hz, 33.0 Hz and 32.0 Hz, respectively.

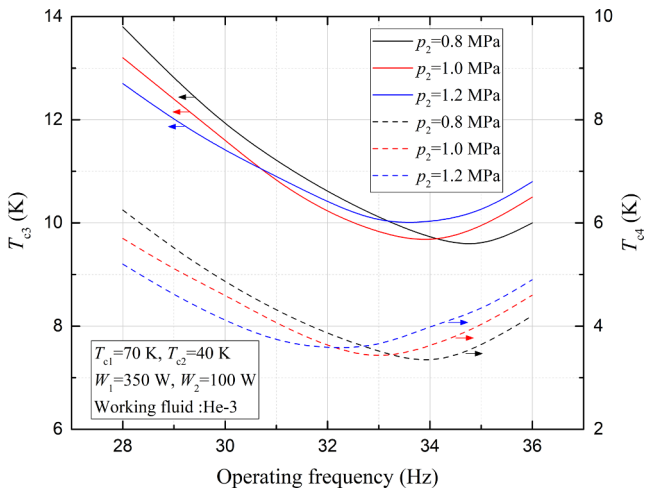


Fig. 15. Variations of T_{c3} and T_{c4} with f_2 and p_2 by using He-3.

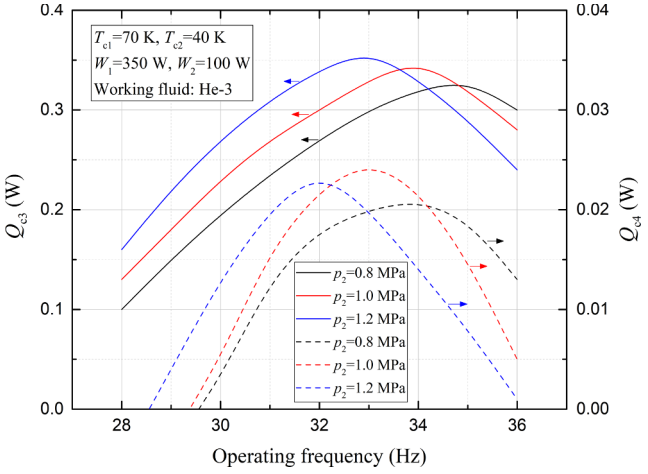


Fig. 16. Variations of Q_{c3} and Q_{c4} with f_2 and p_2 by using He-3.

Fig. 16 shows the variations of Q_{c3} and Q_{c4} with f_2 and p_2 while T_{c3}

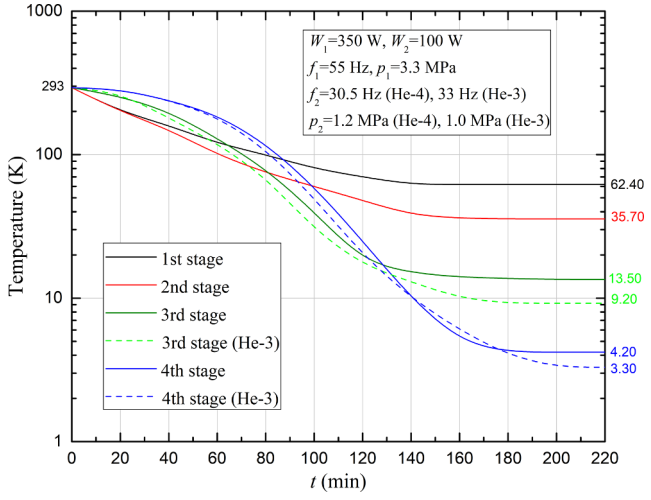


Fig. 17. Cool-down curves of the four stages.

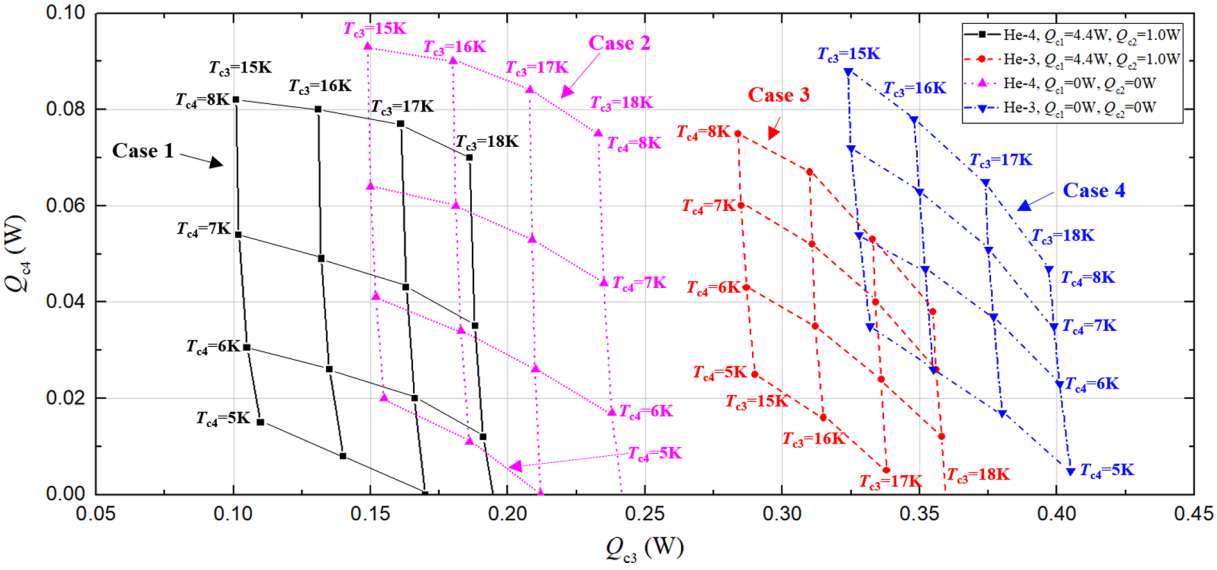


Fig. 18. Temperature-cooling power maps of the third and fourth stages.

and T_{c4} are fixed at 15 K and 5 K, respectively. At the third stage, when p_2 increases from 0.8 MPa to 1.0 MPa and then to 1.2 MPa, the maximum value of Q_{c3} accordingly increases from 0.33 W to 0.35 W and finally to 0.36 W. At the fourth stage, when p_2 is 1.0 MPa and f_2 is 33 Hz, Q_{c4} achieves the maximum value of 0.026 W. Therefore, with He-3 in compressor II, f_2 and p_2 are fixed at 33 Hz and 1.0 MPa, respectively. According to the simulated results in Section 3.2 of Part A [1], the optimal f_2 and p_2 with He-3 are 32 Hz and 1.0 MPa, respectively, which are very close to the experimental ones.

Fig. 17 shows the cool-down curves of the four stages. By using He-4 in the last two stages, the no-load temperatures of the four stages are 62.40 K, 35.70 K, 13.50 K and 4.23 K, respectively. The input powers of compressor I and compressor II are 350 W and 100 W, respectively. While using He-3 to replace He-4 in compressor II, the no-load temperatures of the last two stages reduce further to 9.5 K and 3.30 K, respectively, which is a significantly improvement. Simultaneous cooling powers are achieved by the four-stage SPTC.

Fig. 18 shows the temperature-cooling power maps of both the third and the fourth stages, in which four cases are tested. The cooling powers at the fourth stage from 5 K to 8 K and the associated cooling powers at the third stage from 15 K to 18 K are presented. Comparing Case 1 to Case 2, it is observed that the cooling performance of the third stage is dramatically improved by using He-3 instead of He-4 in the last two stages, while the improvement of the fourth stage is insignificant. Similar conclusions can also be gotten while comparing Case 3 to Case 4. Comparing Case 1 to Case 3, we found that adding heat loads to the first two stages degrades the cooling performances of the last two stages, but the reduction is acceptable. Similar conclusions can also be gotten while comparing Case 2 to Case 4. These results indicate that the four-stage SPTC has the ability of simultaneously providing multiple cooling powers. For example, without cooling powers at the first two stages, the last two stages can simultaneously provide 0.332 W at 15 K and 0.034 W at 5 K, respectively. With cooling powers at the first two stages, the SPTC simultaneously achieves four cooling powers of 4.4 W at 70 K, 1.0 W at 40 K, 0.29 W at 15 K and 0.025 W at 5 K, respectively.

Based on the experimental investigation, the developed four-stage SPTC is proved to be capable of directly reaching a temperature of below the liquid helium, and also of simultaneously providing four cooling powers at four typical temperatures. Satisfactory agreements are observed between theoretical and experimental studies.

3. Conclusions

This paper conducts the experimental verifications of a four-stage SPTC. The actual four-stage SPTC is worked out to verify the theoretical analyses in Part A [1]. Experimental results are discussed and compared with the simulated ones.

A series of experiments are carried out to investigate the interactions among the four stages. The operating frequency, charge pressure and input power of the last two stages are optimized with He-4 and He-3, respectively. The experimental results show that a no-load temperature of 4.2 K is reached with He-4 and it further decreases to 3.3 K with He-3. The ability to providing simultaneous cooling powers is also tested. With He-3, the cooling powers of 4.4 W at 70 K, 1.0 W at 40 K, 0.29 W at 15 K and 0.025 W at 5 K are simultaneously achieved by the four stages, respectively.

Satisfactory agreements are observed between theoretical and experimental studies.

Declaration of Competing Interest

The authors declared that they have no conflicts of interest to this work.

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References

[1] Dang HZ, Zha R, Tan J, et al. Investigations on a 3.3 K four-stage Stirling-type pulse tube cryocooler. Part A: Theoretical analyses and modelling. Cryogenics 2019. https://doi.org/10.1016/j.cryogenics.2019.103014.
[2] Dang HZ. Development of high performance moving-coil linear compressors for space Stirling-type pulse tube cryocoolers. Cryogenics 2015;68:1–18.
[3] Dang HZ, Zhang L, Tan J. Dynamic and thermodynamic characteristics of the moving-coil linear compressor for the pulse tube cryocooler. Part A: Theoretical analyses and modeling. Int J Refrig 2016;69:480–96.